

Introductory
Night Sky
photography

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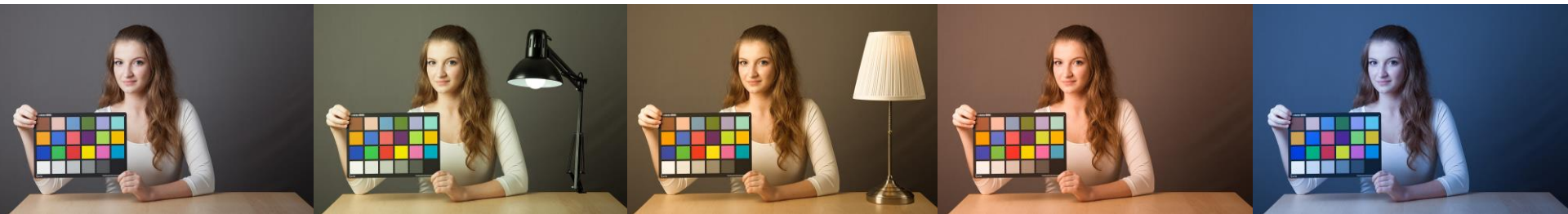
Processing, tracking & Deep sky imaging



Camera settings

White Balance

Set White Balance to daylight/Sun
for the most true-colour rendition.



Images by Alex1ruff via Wikipedia

Camera settings

Additional

Always use a tripod!



Use remote
shooting/control:

- To avoid camera shake
- To take a batch of photos





stellarium

latest version is 24.4



Linux
source



Linux
snap



Linux
amd64;
Qt5;
AppImage



macOS
11.0+;
Qt6;
universal



Windows
x86_32;
Qt5;
Windows
7+



Windows
x86_64;
Qt5;
Windows
7+

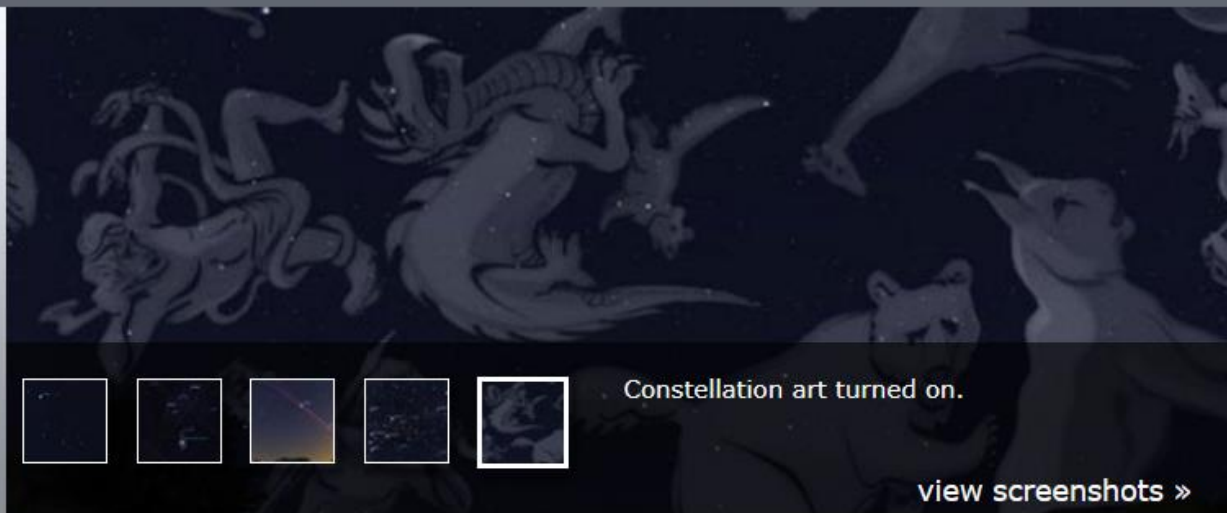


Windows
x86_64;
Qt6;
Windows
10+



Stellarium
Web

Stellarium is a free open source planetarium for your computer. It shows a realistic sky in 3D, just like what you see with the naked eye, binoculars or a telescope.



Constellation art turned on.

[view screenshots »](#)

features

sky

- default catalogue of over 600,000 stars
- extra catalogues with more than 177 million stars
- default catalogue of over 80,000 deep-sky objects
- extra catalogue with more than 1 million deep-sky objects
- asterisms and illustrations of the constellations

news

- [Stellarium 24.4](#)
- [Stellarium 24.3](#)
- [Stellarium 24.2](#)
- [Stellarium 24.1](#)
- [Stellarium 23.4](#)
- [Stellarium 23.3](#)
- [Stellarium 23.2](#)
- [Stellarium 23.1](#)
- [Stellarium 1.2](#)
- [Stellarium 1.1](#)

collaborate

You can learn more about Stellarium, get support and help the project from these links:

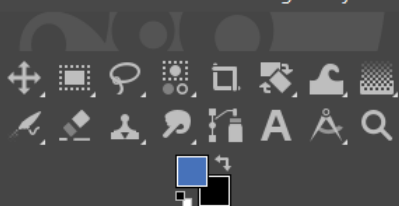
- [discussions](#)
- [mailing list](#)
- [wiki](#)
- [FAQ](#)
- [scripts](#)
- [landscapes](#)

Astrophotography Processing

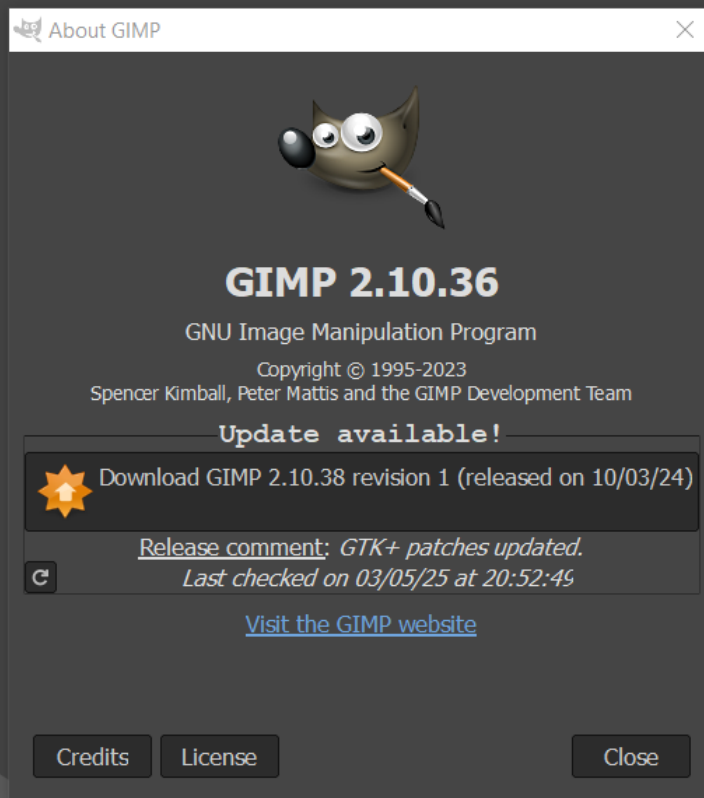
I use:

- Sequator
 - for stacking and aligning images
- GIMP
 - to improve the final image (brightness, contrast, etc)

Both are free windows software.



- | | |
|--|---------------------------------|
|  | Custom |
|  | FG to BG (Hardedge) |
|  | FG to BG (HSV dockwise hue) |
|  | FG to BG (HSV counter-dockwise) |
|  | FG to BG (RGB) |
|  | FG to Transparent |
|  | FG to Transparent (Hardedge) |
|  | Abstract 1 |
|  | Abstract 2 |
|  | Abstract 3 |
|  | Aneurism |
|  | Blinds |
|  | Blue Green |
|  | Browns |
|  | Brushed Aluminium |
|  | Burning Paper |
|  | Burning Transparency |
|  | Caribbean Blues |
|  | CD |
|  | CD Half |
|  | Coffee |
|  | Cold Steel |
|  | Cold Steel 2 |
|  | Crown molding |
|  | Dark 1 |
|  | Deep Sea |
|  | Default |
|  | Flare Glow Angular 1 |
|  | Flare Glow Radial 1 |



Processing steps

1. Take multiple images
2. Calibration
 - Take & subtract a dark frame

Dark frame

ISO 3200

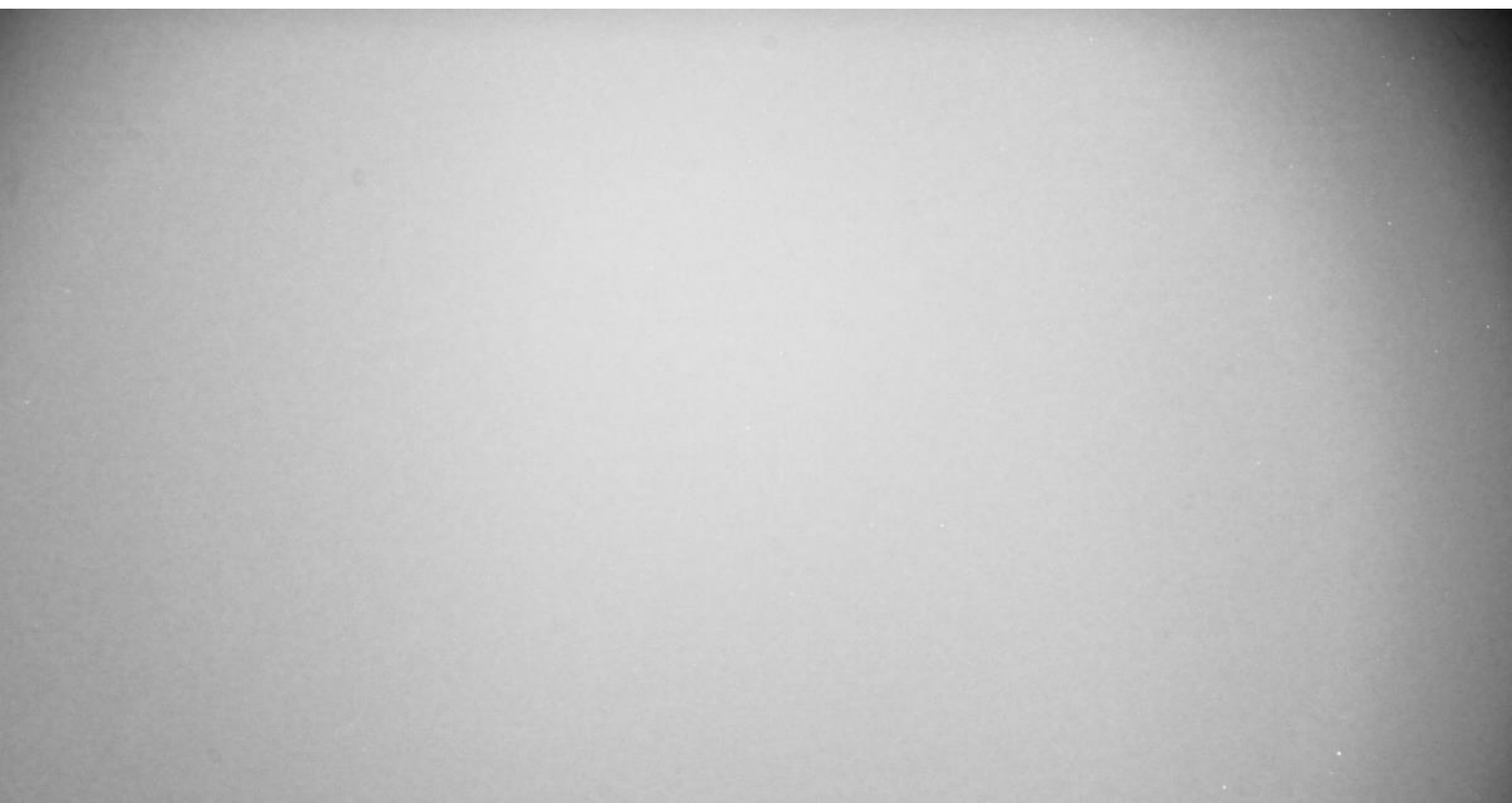
Processing steps

1. Take multiple images

2. Calibration

- Take & subtract a dark frame
- Take & divide by a flat field

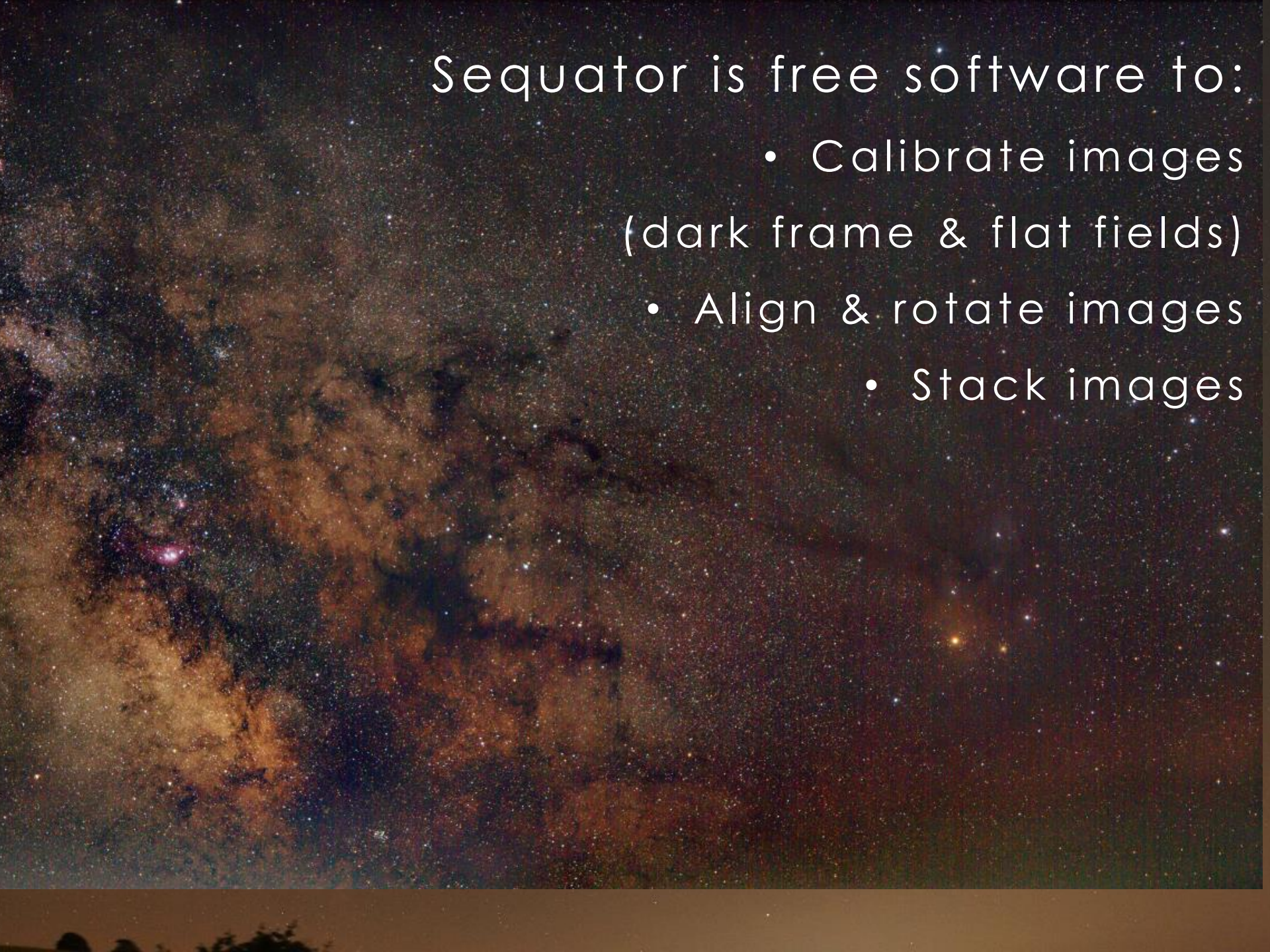
Flat field image



A photograph of a white area shows where more or less light travels through the lens

Processing steps

1. Take multiple images
2. Calibration
 - Take & subtract a dark frame
 - Take & divide by a flat field
3. Align & stack your images
4. Additional processing (brightness, contrast, colours, light pollution reduction, etc...)



Sequator is free software to:

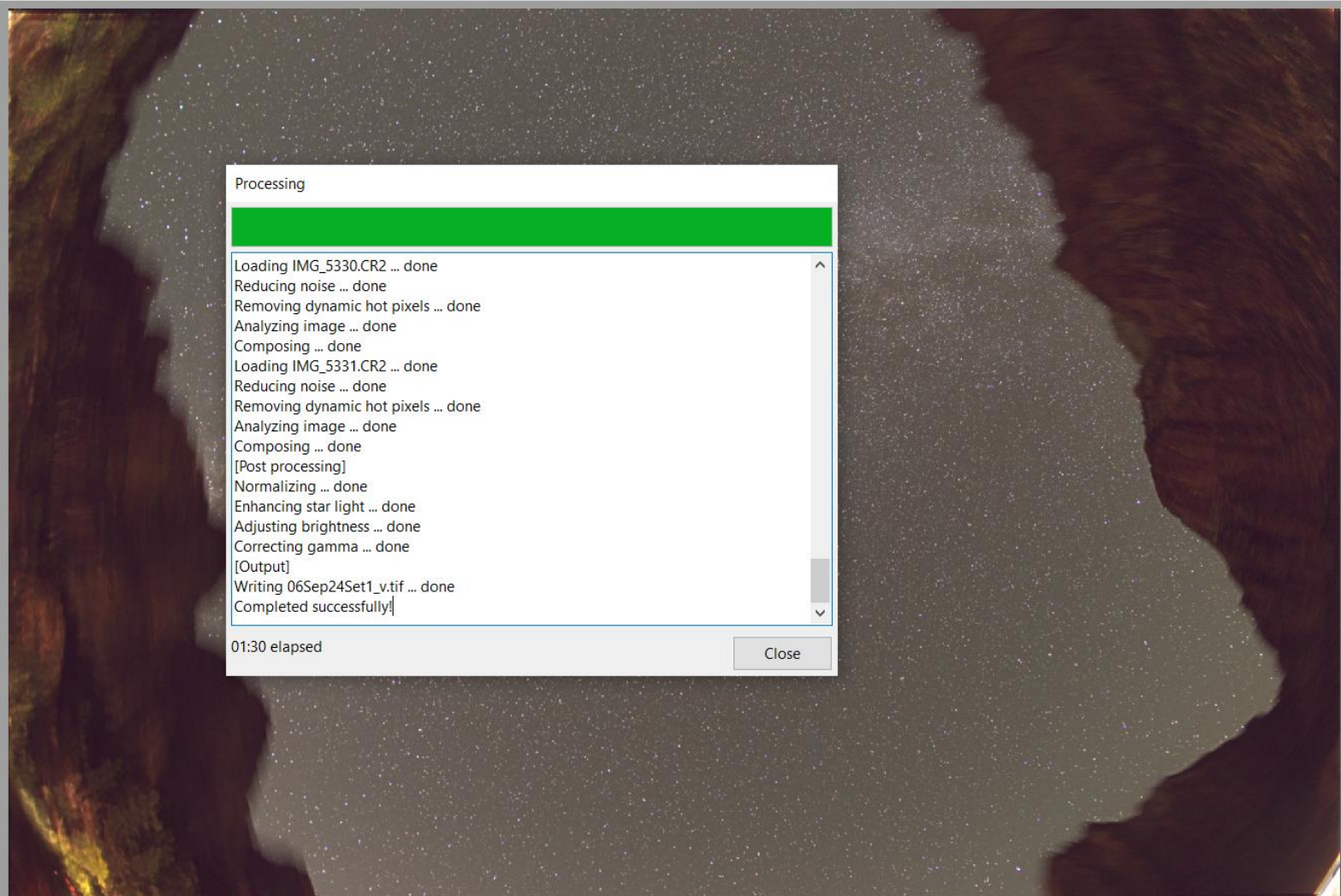
- Calibrate images
(dark frame & flat fields)
- Align & rotate images
- Stack images

- Star images
 - 24 files
- Base image
 - IMG_5319.CR2
- Noise images
 - IMG_5333.CR2
- Vignetting images
- Output
 - 06Sep24Set1_v.tif

- Composition: Align stars
 - Accumulation
- Sky region: Full area
- Auto brightness: On
- High dynamic range: Off
- Remove dynamic noises: On
- Reduce distortions: effects: Complex
- Reduce light pollution: Off
- Enhance star light: On
- Merge 4 pixels: Off
- Time-lapse: Off
- Color space: sRGB

Start

Result: E:\2024PhotosBackup\Arizona2024\astro\06Sep24\06Sep24Set1\06Sep24Set1_v.tif [Open](#)



Processing

Loading IMG_5330.CR2 ... done
Reducing noise ... done
Removing dynamic hot pixels ... done
Analyzing image ... done
Composing ... done
Loading IMG_5331.CR2 ... done
Reducing noise ... done
Removing dynamic hot pixels ... done
Analyzing image ... done
Composing ... done
[Post processing]
Normalizing ... done
Enhancing star light ... done
Adjusting brightness ... done
Correcting gamma ... done
[Output]
Writing 06Sep24Set1_v.tif ... done
Completed successfully

01:30 elapsed

Close

Project File Help

- Star images
 - 24 files
- Base image
 - IMG_5319.CR2
- Noise images
 - IMG_5333.CR2
- Vignetting images
- Output
 - 06Sep24Set1_v.tif

Result: E:\2024PhotosBackup\Arizona2024\astro\06Sep24\06Sep24Set1\06Sep24Set1_v.tif [Open](#)

- ☒ Composition: Align stars
 - Accumulation
- ☐ Sky region: Full area
- ☒ Auto brightness: On
- ☐ High dynamic range: Off
- ☒ Remove dynamic noises: On
- ☒ Reduce distort. effects: Complex
- ☐ Reduce light pollution: Off
- ☒ Enhance star light: On
- ☐ Merge 4 pixels: Off
- ☐ Time-lapse: Off
- ☒ Color space: sRGB

 Start



Double click "Star images" to add image(s)



The Earth rotates...





With a zoom lens, you need an
equatorial Mount to follow the stars



30x 30 second photographs, **stacked**



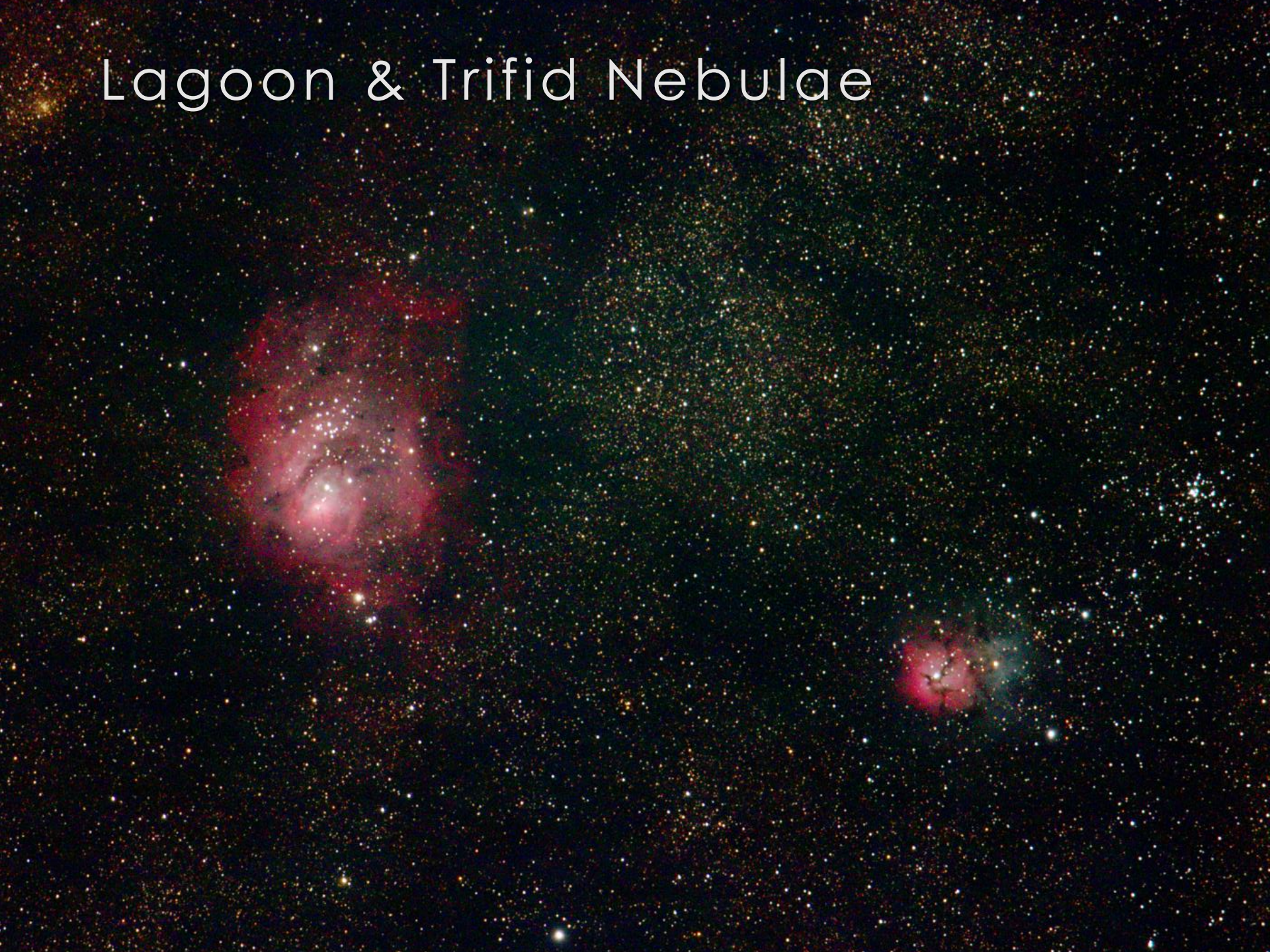
30x 30 second photographs, ***aligned & stacked***

Tracking mounts



Follow the
Stars
Moon
Sun
etc

Lagoon & Trifid Nebulae



The Andromeda Galaxy



Telescopes

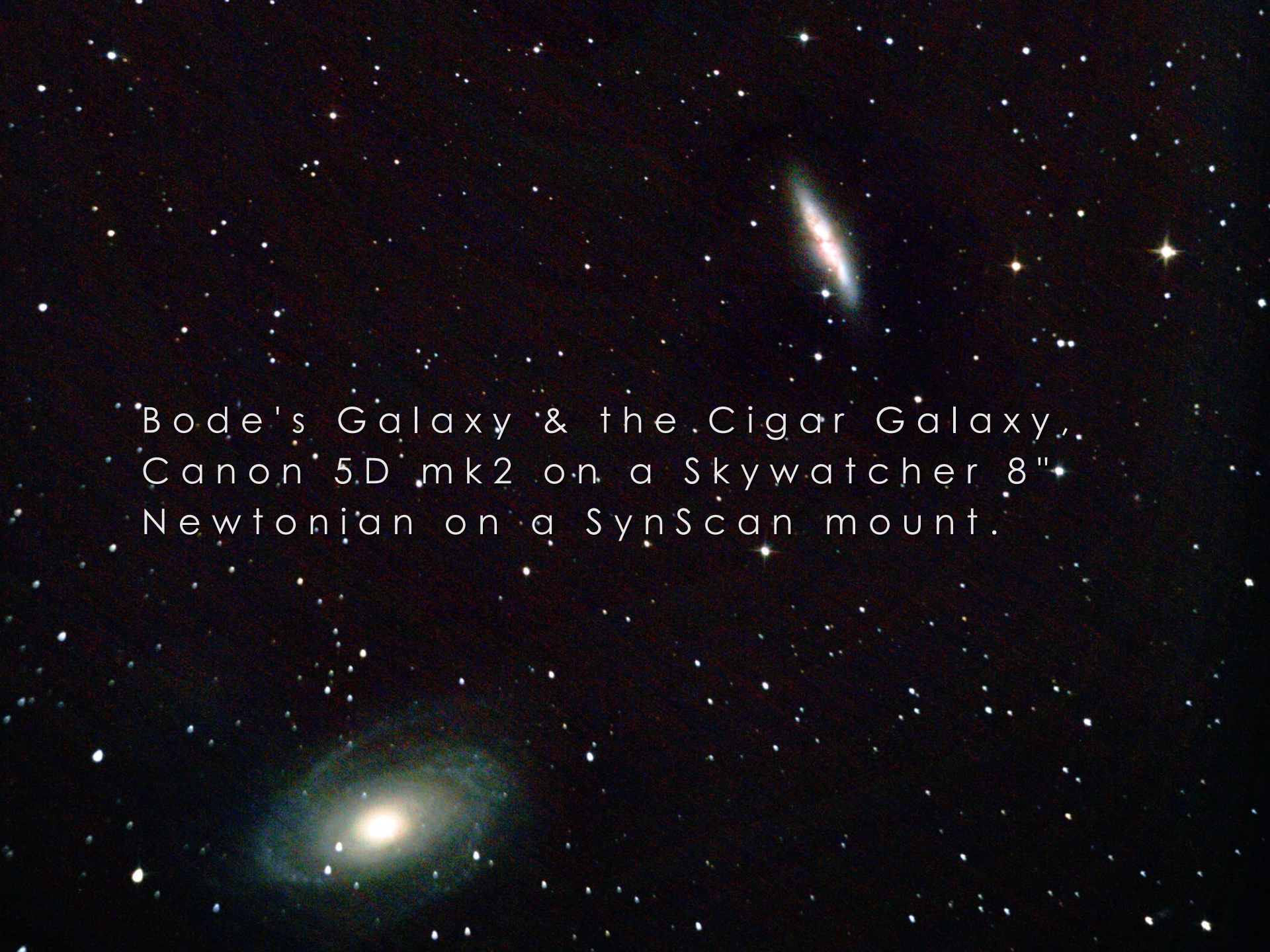


You don't need a
telescope for
astrophotography,
but...

Telescopes



Can get camera
to telescope
adapters...



Bode's Galaxy & the Cigar Galaxy,
Canon 5D mk2 on a Skywatcher 8"
Newtonian on a SynScan mount.



Exposure time

You can
increase the
exposure
time – but
accurate
tracking is
required



Exposure time

You can
increase the
exposure
time – but
accurate
tracking is
required

Processing & Tracking summary

- Brings out more details
- Tracking essential for zooming into individual targets

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